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Project update meeting
Week 7

A TMD-oriented analysis of $\pi^+\pi^-$ pairs in e^+e^- collisions

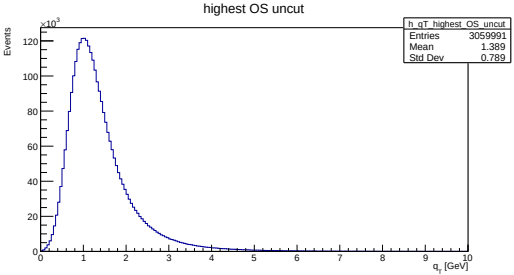
Outline (Week 7)

- Compare **OS** and **SS** event yields with the same thrust-axis setup.
- Apply a **percentage-based cut** to the selected pion.
- Check whether the tighter selection introduces an **OS vs SS bias**.

Uncut sample (15 million events)

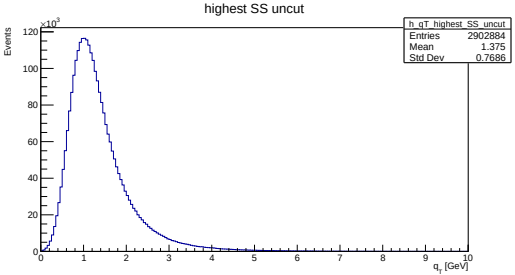
OS

No percentage cut applied.



SS

Baseline reference sample.

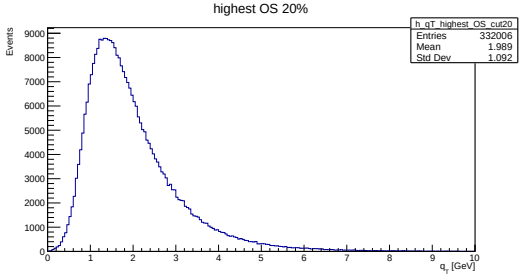


Baseline comparison with **15 million** generated events before any fractional pion cut.

20% pion-fraction cut

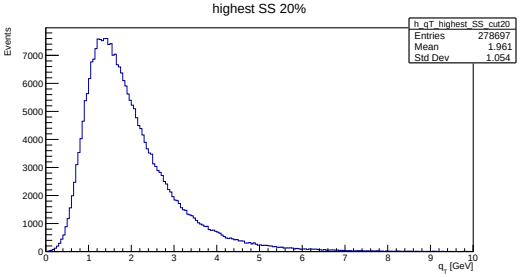
OS

Require the selected pion to pass the 20% cut.



SS

Same cut, same event selection.

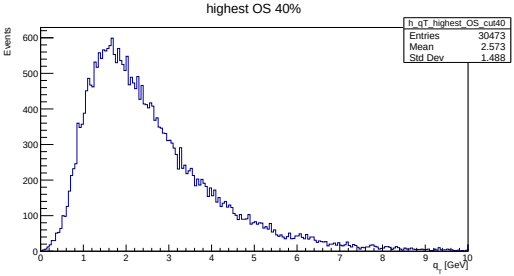


First tightening step: compare whether OS and SS event counts fall at the same rate.

40% pion-fraction cut

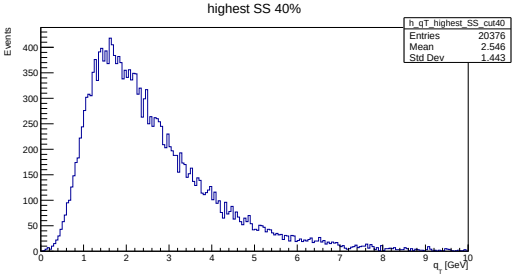
OS

Stronger fractional cut on the selected pion.



SS

Matched SS comparison.

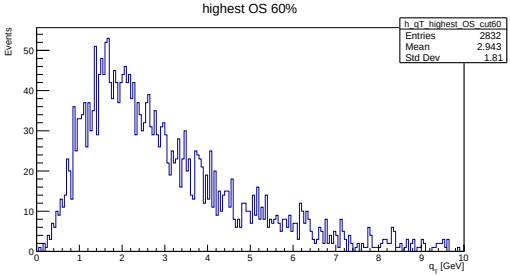


At 40%, any OS/SS difference should start to show more clearly if the cut is biased.

60% pion-fraction cut

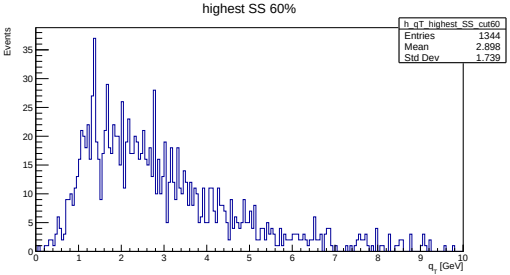
OS

Tight selection.



SS

Same cut applied identically.

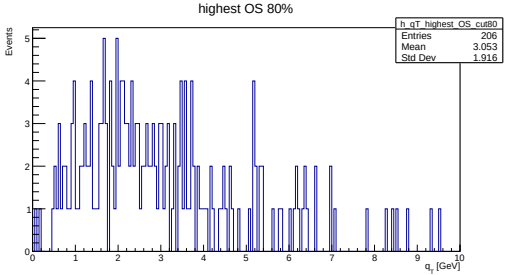


This is the stronger-stress region: if one charge class is affected more, it should be visible here.

80% pion-fraction cut

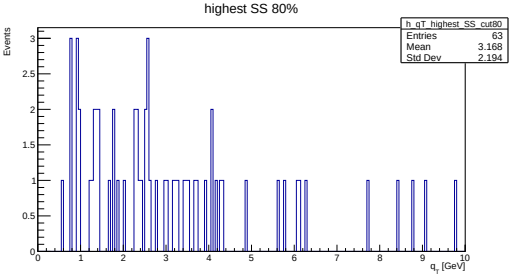
OS

Very tight leading-pion requirement.



SS

Final comparison point.



Final scan point: compare whether OS and SS remain consistent under the tightest cut.

Questions?